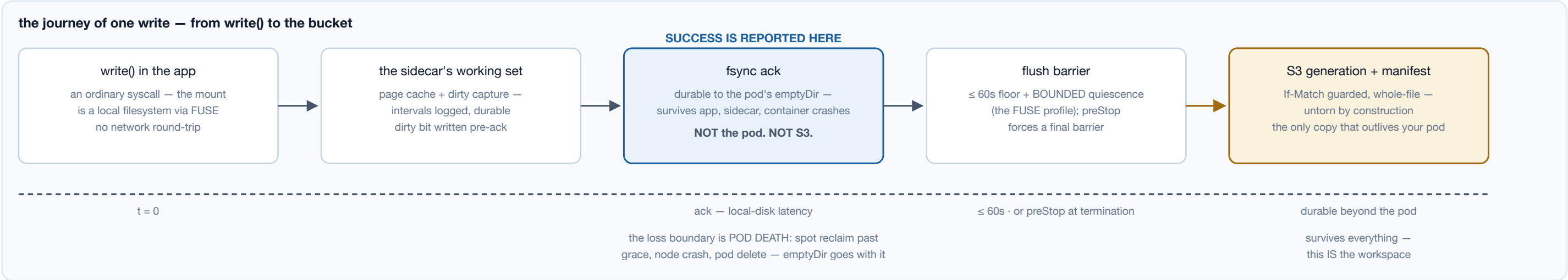


flint-fuse keeps the same write-back tier but moves it into your pod: full local POSIX inside your own subtree, versioned snapshots for everyone else — and the ack’s durability domain shrinks from a crash-surviving PVC to your pod's emptyDir, which lives exactly as long as your pod does.



who sees your write, and when	
you — your pod, your subtree	immediately — local POSIX at disk speed
other containers in your pod	immediately — the same mount (HostToContainer)
another pod reading your subtree	after YOUR next flush + THEIR next revalidate — a consistent snapshot at most one flush interval + one revalidation old; never read-your-peers'-writes
the project service — S3 browser	after your flush barrier — untorn whole-file generations; your subtree manifest (seq) is the snapshot
your replacement pod — clean death	everything — preStop drains: flush barrier, tombstones, lease release (~17s)
your replacement pod — unclean	only what was PUBLISHED — plus a 60–110s subtree write-lockout while the dead lease ages out · cross-pod O_EXCL create: strongly consistent (sync PUT)

not guaranteed — say these out loud

- a successful write is NOT in S3 — and no PVC stands behind you.**
fsync means your pod's emptyDir. The RPO is per POD, and on a spot fleet, pod death is routine — not a disaster scenario.
- your hottest files are your most at-risk files.**
a WAL or log written continuously defers under quiescence; the FUSE profile BOUNDS the deferral and preStop force-flushes — a hard node crash still drains nothing.
- no cross-pod locks, no shared sqlite / git — refused, not discouraged.**
ENOLCK and EXDEV at the FUSE layer; those workloads route to the hub front end, which remains the coherence authority.
- S3-console writes to your live subtree are out of contract.**
same three fates as hub mode — destroyed, undetected, or adopted; the project role is read-only for a reason.

drain arithmetic: the 120s spot notice publishes ~9–15 GiB at the measured 8–13 s/GiB; the K8s default 30s grace ~2–3 GiB; a hard crash, nothing.

The contract in one sentence: inside your subtree, flint-fuse is a local filesystem — lower write latency than NFS, the same write-back posture toward S3 — but the ack’s durability domain shrinks from a crash-surviving PVC to a pod-lifetime emptyDir, so the RPO stops being a disaster bound and becomes the everyday loss bound for unclean pod death. Everyone else reads your workspace as versioned snapshots: S3-strong per object, RPO-stale per subtree, revalidated against the manifest ETag. Three consistency upgrades over plain S3-FUSE gateways: writes are locally full POSIX (in-place, rename, private mmap), publishes are CAS-guarded whole-file images that are never torn, and a create can be made globally exclusive on request (O_EXCL as a synchronous If-None-Match:* PUT — the one strongly consistent cross-pod primitive in this mode). Two demotions from hub mode, stated plainly: there is no cross-pod coherence authority, and there is no durable ground under an acked write. Size `terminationGracePeriodSeconds` and the dirty-set cap together, run an interruption handler, and treat the flush profile as what it now is — a durability contract, not a cost knob.